

Practice Exercises (Bayes Rule)

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Don't worry if you find these problems hard. It's natural to feel confused the first time you work through new concepts, and struggling with a problem is part of how you learn it. This is just practice. You'll all be fine. :)

Exercise 1. A doctor is called to see a sick child. The doctor has prior information that 90% of sick children in that neighborhood have the flu, while the other 10% are sick with measles. Let F stand for the event of a child being sick with flu and M stand for the event of a child being sick with measles. Assume for simplicity that $F \cup M = \Omega$, i.e., that there are no other maladies in that neighborhood.

A well-known symptom of measles is a rash (the event of having which we denote R). Assume that the probability of having a rash if one has measles is $P(R | M) = 0.95$. However, occasionally children with flu also develop a rash, and the probability of having a rash if one has flu is $P(R | F) = 0.08$.

Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

Solution.

We use Bayes's formula.

$$\begin{aligned} P(M | R) &= \frac{P(R | M) P(M)}{P(R | M) P(M) + P(R | F) P(F)} \\ &= \frac{0.95 \times 0.10}{0.95 \times 0.10 + 0.08 \times 0.90} \simeq 0.57. \end{aligned}$$

This is nowhere close to the 95% of $P(R | M)$.

Exercise 2. It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?

Solution.

Define the events

A = event that an email is detected as spam,

B = event that an email is spam,

B^c = event that an email is not spam.

We know

$$P(B) = P(B^c) = 0.5, \quad P(A | B) = 0.99, \quad P(A | B^c) = 0.05.$$

Hence, by Bayes's formula, we have

$$\begin{aligned} P(B^c | A) &= \frac{P(A | B^c) P(B^c)}{P(A | B) P(B) + P(A | B^c) P(B^c)} \\ &= \frac{0.05 \times 0.5}{0.99 \times 0.5 + 0.05 \times 0.5} \\ &= \frac{5}{104} \approx 0.048. \end{aligned}$$

Exercise 3. In a study, physicians were asked what the odds of breast cancer would be in a woman who was initially thought to have a 1% risk of cancer but who ended up with a positive mammogram result (a mammogram accurately classifies about 80% of cancerous tumors and 90% of benign tumors.)

95 out of a hundred physicians estimated the probability of cancer to be about 75%. Do you agree?

Solution.

Introduce the events:

$+$ = mammogram result is positive,
 B = tumor is benign,
 M = tumor is malignant.

Note that $B^c = M$. We are given $P(M) = 0.01$, so $P(B) = 1 - P(M) = 0.99$. We are also given the conditional probabilities $P(+ | M) = 0.80$ and $P(- | B) = 0.90$, where the event $-$ is the complement of $+$. Thus $P(+ | B) = 1 - P(- | B) = 0.10$.

Bayes's formula in this case is

$$\begin{aligned} P(M | +) &= \frac{P(+ | M) P(M)}{P(+ | M) P(M) + P(+ | B) P(B)} \\ &= \frac{0.80 \times 0.01}{0.80 \times 0.01 + 0.10 \times 0.99} \\ &\simeq 0.075. \end{aligned}$$

So the chance would be 7.5%, a far cry from the common estimate of 75%.